INSIDE THE SYSTEM

FROM PIXELS TO MOTION.

One webcam image becomes five finger controls. Here is the path the data takes.

THE LIVE HAND-TRACKING PIPELINE

01 · INPUT
Webcam
Captures a mirrored video frame. Python requests 640 × 480.

02 · VISION
MediaPipe
Finds 21 estimated landmarks on one visible hand.

03 · MATH
Finger curl
Vector angles are mapped to 0–90 curl commands.

04 · GATE
Safe check
Browser twin blocks a flagged pose or sudden wrist jerk.

05 · OUTPUT
Two paths
Three.js twin on screen, or Python serial → Arduino servos.

How does code estimate a bend?

For a finger, the program uses three landmark points. Two vectors meet at the middle joint.



$$\theta = \cos^{-1}\left(u \cdot v \,/\, |u|\,|v| \right)$$

A straighter finger is near 180°. The code maps a chosen straight-to-curled range to a control value. It is an estimate from the camera, not a real joint-angle measurement.

What the numbers mean

21 points
MediaPipe hand landmarks

5 channels
Thumb, index, middle, ring, pinky

0 → 90
Open-to-curled command values from Python

25% smoothing
Arduino eases toward each new target

The serial bridge sends packets like `<45,90,90,85,70>` at 115,200 baud.

Safety layer

In the browser demo: a raised-middle-finger pose or sudden wrist jerk turns the check red and holds the robot at its last safe pose. The hand meters still show the unfiltered estimates.

Science fair measurements

The recorder collects 100 detected-hand results with processing delay, frame rate, finger-curl estimates, camera distance, and room-light input. Change one condition at a time and compare your own trials.

Important distinction: The browser safety gate and the Python/Arduino motor path are separate. The Python motor path does not currently include that same gesture/jerk block. The web pickup is simulated; only the optional maker build moves physical servos.